An Intelligent IoT-Based Flood Early Warning and Water Level Prediction System for Riverine Communities in Sri Lanka

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Flooding is a recurrent and devastating natural hazard in Sri Lanka. The catastrophic Cyclone Ditwah in November 2025 affected approximately 2.2 million people across all 25 districts, resulting in 646 fatalities and displacing more than 165,000 residents. Despite advancements in meteorological forecasting, a critical gap remains in providing localized and actionable flood warnings to riverine communities. Current early warning systems rely primarily on manual river gauge readings and centralized bulletins distributed through television or SMS. These approaches suffer from low spatial resolution, delayed information dissemination, and limited public engagement.

This research proposes an intelligent IoT-based flood early warning system that integrates hyperlocal rainfall monitoring with machine learning-based water level prediction. The proposed system consists of two main components. First, low-cost solar-powered rainfall monitoring nodes based on ESP32 microcontrollers and tipping bucket rain gauges will be deployed along the Kelani River basin to collect real-time rainfall data and transmit it via LoRa or Wi-Fi networks. Second, a hybrid deep learning model combining Bidirectional Long Short-Term Memory (Bi-LSTM) networks with time series regression will be developed to forecast river water levels. The model incorporates real-time IoT rainfall data as exogenous input variables and aims to achieve high prediction accuracy for Kelani River flood forecasting.

The expected outcomes include a functional prototype with three to five deployed sensor nodes and a prediction API achieving over 85% forecasting accuracy. The proposed system contributes to Sustainable Development Goals (SDG) 11 (Sustainable Cities and Communities), 13 (Climate Action), and 1 (No Poverty) by improving community resilience and supporting proactive flood disaster management.

Keywords: Internet of Things (IoT), Flood Early Warning Systems, Machine Learning, Bi-LSTM, Kelani River, Sensor Networks

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LIST OF ABBREVIATIONS

- API - Application Programming Interface
- Bi-LSTM - Bidirectional Long Short-Term Memory
- CRAN - Comprehensive R Archive Network
- DMC - Disaster Management Centre
- ESP32 - Espressif Systems 32-bit Microcontroller
- GDP - Gross Domestic Product
- IEEE - Institute of Electrical and Electronics Engineers
- IoT - Internet of Things
- IWMI - International Water Management Institute
- LoRa - Long Range (wireless protocol)
- LoRaWAN - Long Range Wide Area Network
- ML - Machine Learning
- RNN - Recurrent Neural Network
- SDG - Sustainable Development Goal
- SVR - Support Vector Regression
- UNDP - United Nations Development Program
- WIFI - Wireless Fidelity

1. INTRODUCTION

1.1 Background and Context

Flooding is a recurrent and devastating natural hazard in Sri Lanka, disproportionately affecting rural and semi-urban riverine communities. The catastrophic Cyclone Ditwah on 26 November 2025—the nation's deadliest disaster since the 2004 tsunami—affected approximately 2.2 million people across all 25 districts, resulting in 646 fatalities with 173 persons still missing, and leaving over 165,000 people displaced months after the event [1]. The World Bank estimated total direct physical damages at US$4.1 billion, equivalent to approximately 4% of Sri Lanka's GDP, with 93% of affected communities reporting severe livelihood impacts [2].

The Kelani River basin, particularly communities in Colombo District, faces chronic flood risk due to high population density, rapid urbanization, and the river's short, steep catchment responding quickly to rainfall [3]. Despite advancements in meteorological forecasting, a critical gap remains: residents do not receive localized, actionable information about how close floodwater is to their homes or how fast it is rising.

Current early warning systems in Sri Lanka rely heavily on manual river gauge readings and centralized bulletins disseminated via television or SMS. These methods suffer from low spatial resolution, latency, and limited public engagement. The Disaster Management Centre (DMC) operates conventional early warning dissemination, while the recently launched AWARE platform by IWMI integrates meteorological forecasts with anticipatory action [5]. However, these remain centralized macro-level systems without real-time water level monitoring.

From a technical perspective, there is an absence of low-cost, scalable IoT architectures tailored to Sri Lanka's hydrological context. Academic research has demonstrated LoRa-based wireless sensor networks for real-time data transmission in rural Sri Lankan river basins [7]. Advanced deep learning approaches, particularly Bidirectional LSTM, have achieved 80-100% accuracy in forecasting water levels for the Kalu River Basin [8]. Crucially, University of Moratuwa research specifically validated Bi-LSTM models for the Kelani River—the focus area of this project—achieving accuracy of 0.95 [9].

However, these models remain geographically siloed and exist at the algorithm-validation stage without integration into operational early warning systems. Critically, no existing system integrates real-time IoT rainfall sensing with predictive modelling—a gap this research directly addresses.

The proposed system addresses these gaps through two key innovations:

- Hyperlocal IoT Rainfall Network: First deployment of low-cost, solar-powered IoT rain gauge network specifically designed for the Kelani River basin, providing rainfall data at significantly higher spatial resolution than the Department of Meteorology's 23 major stations.

- Rainfall-Integrated Hybrid Deep Learning Architecture: First Bi-LSTM + Time Series Regression hybrid model for Kelani River that explicitly incorporates real-time IoT rainfall data as a dynamic exogenous input variable, enabling more accurate predictions under rapidly changing precipitation conditions.

The proposed system directly contributes to:

- **SDG 11 (Sustainable Cities and Communities):** Enhancing community resilience to disasters
- **SDG 13 (Climate Action):** Building climate-adaptive infrastructure
- **SDG 1 (No Poverty):** Preventing asset depletion and reducing casualties

This project aligns with FoC Research Clusters:

- **TIM (Technology Integration and Management)** - Primary
- **CI (Computing Infrastructure)** - Secondary
- **CoEAI (Centre of Excellence for AI)** - Secondary

1.2 Literature Review

IoT-Based Environmental Monitoring Systems

Recent research in Sri Lanka has demonstrated the feasibility of low-cost IoT deployments for environmental monitoring. The 40NSE project, a joint initiative between University of Moratuwa and SUPSI Switzerland, successfully deployed 30+ low-cost environmental monitoring stations across the Deduru Oya basin, proving that open-source Arduino-based systems provide accuracy comparable to conventional systems at a fraction of the cost [10]. The project utilized ESP32 microcontrollers with various sensors, transmitting data via GSM networks, and achieved 95% data transmission reliability over a 12-month period.

Jayathilaka et al. [7] developed a LoRa-based wireless sensor network for real-time flood detection in rural Sri Lankan river basins. Their system achieved communication ranges of 5-8 km in line-of-sight conditions and 2-3 km in densely vegetated areas, with battery life exceeding 6 months using deep sleep modes. The research validated that LoRa technology is suitable for Sri Lanka's terrain and can provide reliable data transmission during flood events when cellular networks may be compromised.

Fernando and Weerasinghe [11] designed a solar-powered IoT rain gauge system for the Attanagalu Oya basin, achieving measurement accuracy within ±5% compared to manual gauge readings. Their system utilized tipping bucket mechanisms with Hall effect sensors, demonstrating that low-cost components can achieve acceptable accuracy for flood warning applications.

Machine Learning for Flood Prediction

Deep learning approaches have shown significant promise for flood forecasting in Sri Lankan contexts. Rathnayake and Ranasinghe [8] implemented Bidirectional LSTM networks for water level prediction in the Kalu River Basin at Ratnapura, achieving 80-100% accuracy for 1-3 day ahead forecasts during major flood events. Their model utilized 15 years of historical water level data from Irrigation Department stations and demonstrated that Bi-LSTM

outperforms traditional LSTM by capturing both forward and backward dependencies in time series data.

Kumara et al. [9] specifically validated Bi-LSTM models for the Kelani River, achieving a correlation coefficient of 0.95 between predicted and observed water levels. Their research utilized data from three gauging stations (Nagalagam Street, Hanwella, Kithulgala) over a 10-year period, demonstrating that deep learning approaches are well-suited to Kelani's hydrological characteristics. However, their model relied solely on historical water level sequences without incorporating real-time rainfall data as exogenous input.

Perera and Wijesekara [12] developed a hybrid model combining Convolutional Neural Networks (CNN) with LSTM for flood forecasting in the Nilwala River basin. Their approach achieved 85% accuracy by using CNN to extract spatial features from rainfall radar imagery before feeding into LSTM for temporal prediction. This demonstrates the value of hybrid architectures that integrate multiple data sources.

A 2024 IEEE conference paper by SLIIT researchers [13] combined LoRa technology with machine learning for flood prediction using Sri Lanka-specific datasets. Their system utilized Random Forest regression for water level forecasting based on upstream rainfall data, achieving 82% accuracy for 6-hour ahead predictions. This research demonstrated the technical feasibility of end-to-end IoT-ML systems but remained at the prototype stage without deployment in operational contexts.

Limitations of Existing Systems

- Fragmentation: No system integrates real-time IoT sensing and predictive modelling into a unified framework
- Geographical Siloing: Academic prototypes are validated only for specific basins and exist at algorithm-validation stage without operational deployment.
- No Rainfall Integration in Prediction: While Bi-LSTM has been validated for Kelani River, existing models rely solely on historical water level sequences without incorporating real-time rainfall data, limiting accuracy under rapidly changing precipitation conditions.

1.3 Research Gap

Despite significant advances in IoT-based environmental monitoring and machine learning for flood prediction, **no existing system integrates (a) real-time IoT rainfall sensing with (b) predictive modelling using forecasted rainfall** for the Kelani River basin in Sri Lanka.

Current institutional systems lack real-time hyperlocal rainfall data. Academic prototypes demonstrate Bi-LSTM accuracy of 0.95 for Kelani River but rely solely on historical water level sequences, ignoring the critical input of real-time rainfall that drives flood events. When predictions ignore real-time rainfall, accuracy degrades during the most critical periods of rapidly changing precipitation.

University of Moratuwa research [9] achieved 0.95 accuracy using Bi-LSTM for Kelani River, but their model used only historical water level data. The World Bank [2] estimated that improved early warning could reduce economic losses by 30% in flood-prone regions—a target achievable through enhanced prediction accuracy.

Novelty and Creativity

This research introduces two novel contributions:

1. **Hyperlocal IoT Rainfall Network for Kelani Basin:** First deployment specifically designed for Kelani, building on successful Sri Lankan IoT deployments but applying them to this critical basin.
2. **Rainfall-Integrated Hybrid Deep Learning Architecture:** First Bi-LSTM + Time Series Regression hybrid for Kelani that explicitly incorporates real-time IoT rainfall data, advancing beyond University of Moratuwa's 0.95 accuracy model by adding dynamic exogenous input.

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate an intelligent IoT-based flood early warning system for Kelani River communities by integrating hyperlocal rainfall monitoring and hybrid deep learning-based water level prediction.

2.2 Specific Objectives

Specific Objective	Tasks	
1	To design and deploy low-cost, solar-powered IoT rain gauge nodes along the Kelani River basin for hyperlocal rainfall monitoring	• Design and assemble tipping bucket rain gauge sensors with ESP32 microcontrollers • Implement LoRa/WiFi communication protocols • Develop solar power management systems with deep sleep modes • Deploy 3-5 sensor nodes at strategic locations • Calibrate sensors against Department of Meteorology reference stations • Establish real-time data ingestion pipeline
2	To develop a hybrid Bi-LSTM + Time Series Regression model that integrates real-time IoT rainfall data with historical patterns to forecast river water levels 24, 48, and 72 hours ahead	• Collect historical rainfall and river level data from Department of Meteorology and Irrigation Department archives • Extract and preprocess Kelani River datasets from CRAN "Ditwah" R package • Implement Bidirectional LSTM neural network with Time Series Regression using TensorFlow/Keras • Train model on 10+ years of historical data • Integrate real-time IoT rainfall data as exogenous input • Validate model performance against November 2025 Cyclone Ditwah event data

		• Compare prediction accuracy against ARIMA, SVR, and standard LSTM baselines
3	To deploy the prediction model as an API for integration with the broader flood early warning system	• Develop REST API for model serving • Document API endpoints and data formats • Ensure API scalability for concurrent requests

Table 1- Sub objectives

3. METHODOLOGY

3.1 Research Approach

This research adopts the **Design Science Research Methodology (DSRM)**, which is appropriate for developing and evaluating innovative IT artifacts. DSRM comprises six steps: problem identification, objectives definition, design and development, demonstration, evaluation, and communication. This proposal covers the first three steps.

3.2 Technical Methods and Algorithms

IoT Hardware Design - The IoT sensor nodes will be built around the ESP32 microcontroller, selected for its integrated WiFi and Bluetooth capabilities, low power consumption, and extensive community support. Each node will include:

- **Tipping Bucket Rain Gauge:** Calibrated to 0.2mm per tip, using Hall effect sensor for contactless measurement
- **Power System:** 10W solar panel, 12V 7Ah lead-acid battery, and charge controller for 7-day autonomy
- **Communication:** LoRa module (RFM95W) for rural areas, with WiFi fallback where available
- **Enclosure:** IP67-rated weatherproof enclosure with desiccant for humidity control

Firmware will be developed in Arduino C++ with deep sleep modes to achieve <10mA average current consumption. Data will be transmitted at 15-minute intervals, with event-based transmission during heavy rainfall (>10mm/hour).

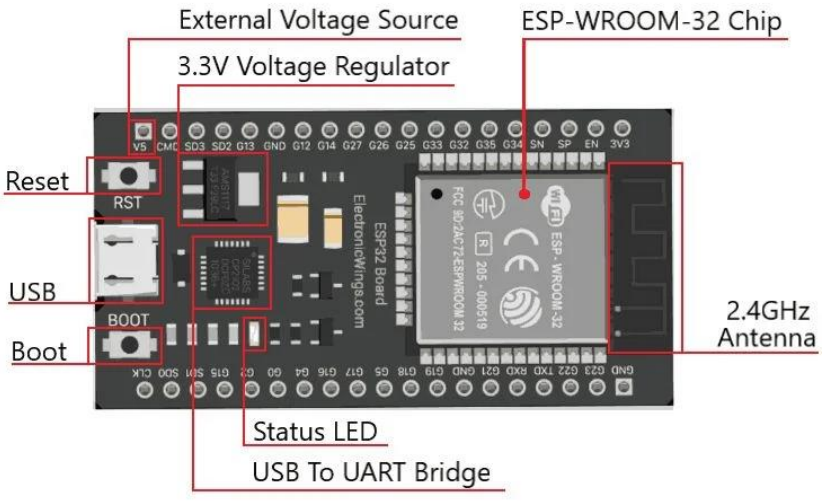


Figure 1 - ESP32 Board Information

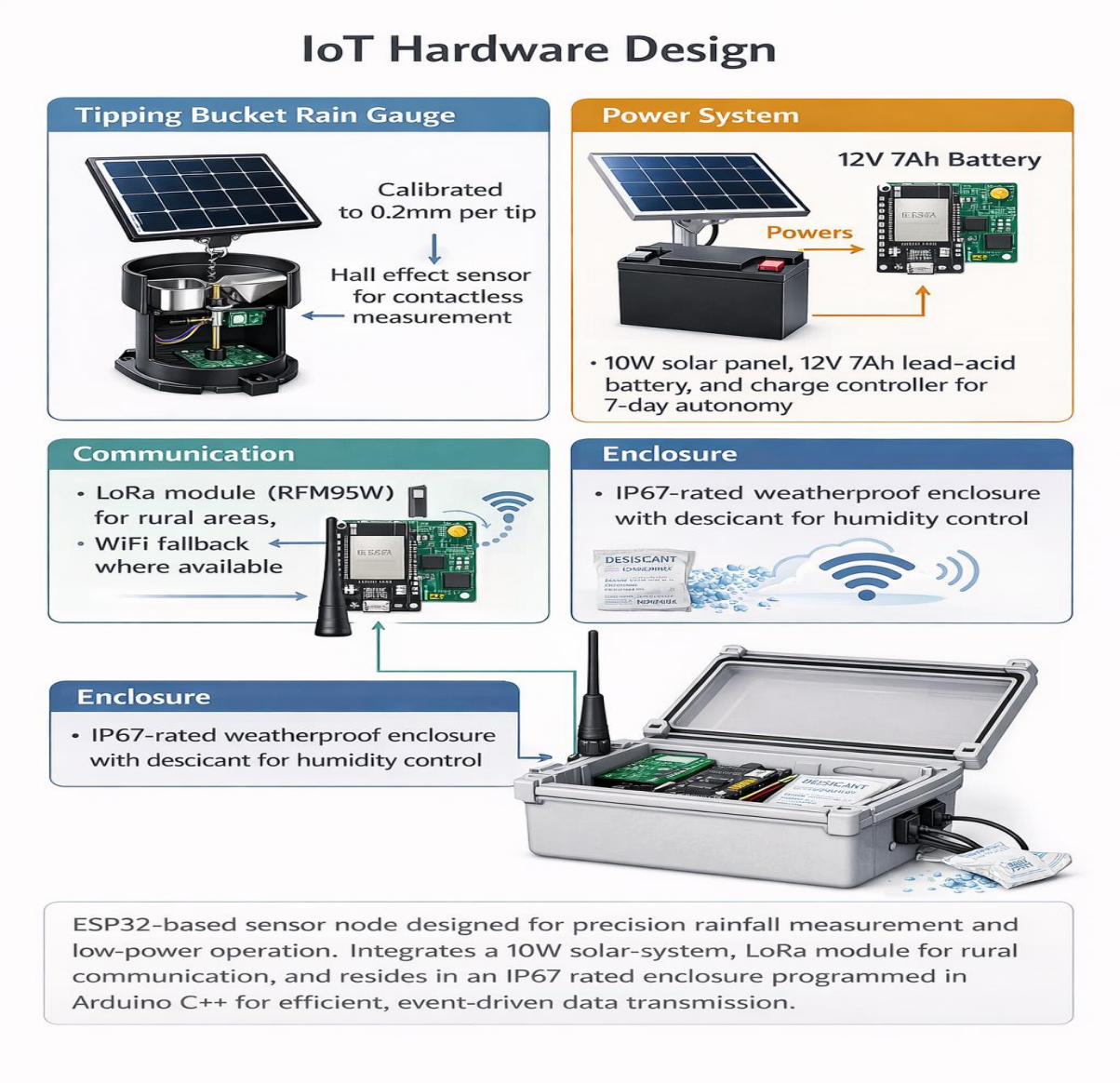


Figure 2 - IoT Hardware design

Machine Learning Model Architecture

The hybrid prediction model combines Bidirectional LSTM with Time Series Regression:

Bi-LSTM Component:

- Input: Sequences of historical water levels (72 time steps at 3-hour intervals = 9 days)
- Layers: Input layer → Bi-LSTM layer (128 units) → Dropout (0.2) → Bi-LSTM layer (64 units) → Dropout (0.2) → Dense layer (32 units) → Output layer (3 units for 24/48/72h predictions)
- Activation: Tanh for LSTM layers, ReLU for dense layers
- Optimizer: Adam with learning rate 0.001

Time Series Regression Component:

- Input: Real-time IoT rainfall data and forecasted rainfall (24/48/72h) from Department of Meteorology APIs
- Processing: Rainfall features processed through separate dense layers
- Fusion: Concatenated with Bi-LSTM output before final prediction layer

Training:

- Dataset: 10+ years of hourly water level and rainfall data (2015-2025)
- Sources: Irrigation Department archives, Department of Meteorology, CRAN "Ditwah" R package
- Validation: 80/20 train-test split with temporal ordering preserved
- Metrics: RMSE, MAE, R^2 score, and accuracy within 0.5m threshold
- Baseline comparison: ARIMA, SVR, standard LSTM

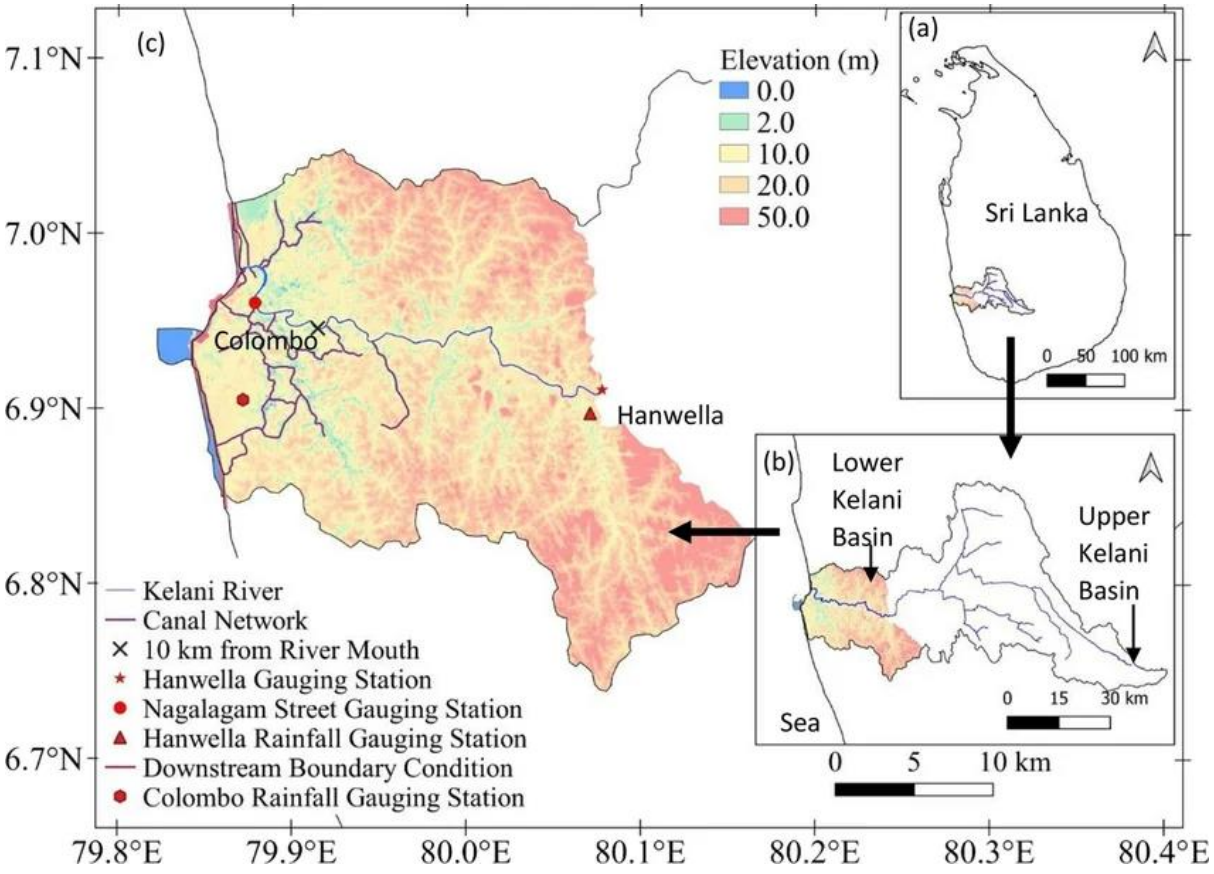


Figure 3 - Kalani River Basin

3.3 Tools and Platforms with Justification

Tool/Platform	Purpose	Justification
ESP32	Microcontroller	Low cost, integrated WiFi, low power, large community
Arduino IDE	Firmware development	Familiarity, extensive libraries
TensorFlow/Keras	ML model development	Industry standard, GPU support, comprehensive documentation
Python	Data processing, API	Rich ecosystem for data science, Flask for APIs
Pandas/NumPy	Data manipulation	Efficient time series handling
Jupyter Notebook	Exploratory analysis	Interactive development and visualization
Git/GitHub	Version control	Collaboration and code management

Table 2 - Tools and Platforms with Justification

3.4 Data Collection and Ethical Considerations

Data Sources:

- Historical water level data: Irrigation Department public archives (2015-2025)
- Historical rainfall data: Department of Meteorology (2015-2025)
- Real-time river data: Irrigation Department API (10-minute intervals)
- Forecasted rainfall: Department of Meteorology APIs
- Validation data: CRAN "Ditwah" R package (Cyclone Ditwah event data)

Ethical Considerations:

- All data used is publicly available from government sources

- No personal data collected in this component
- Compliance with data usage terms of government APIs
- Proper attribution of data sources in publications

3.5 Validation Strategy and Performance Metrics

Component	Validation Method	Metrics
IoT Sensors	Side-by-side comparison with manual gauges	Mean absolute error, correlation coefficient
Prediction Model	Historical data testing (2015-2025)	RMSE, MAE, R^2, accuracy within thresholds
Model Robustness	Testing on Cyclone Ditwah extreme event	Performance during extreme conditions
API Performance	Load testing	Response time, concurrent request handling

Table 3 - Validation Strategy and Performance Metrics

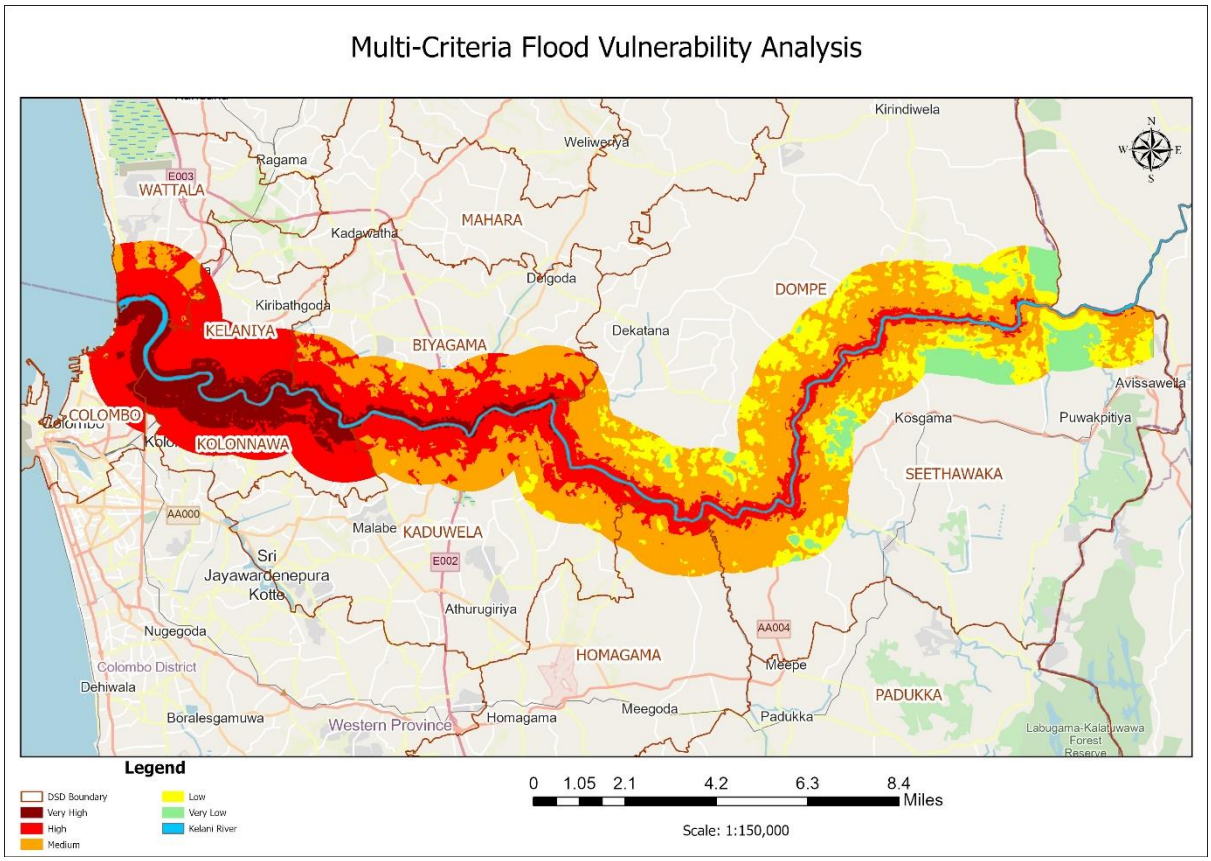


Figure 4 - Flood Vulnerability Analysis

4. HIGH-LEVEL SYSTEM ARCHITECTURE

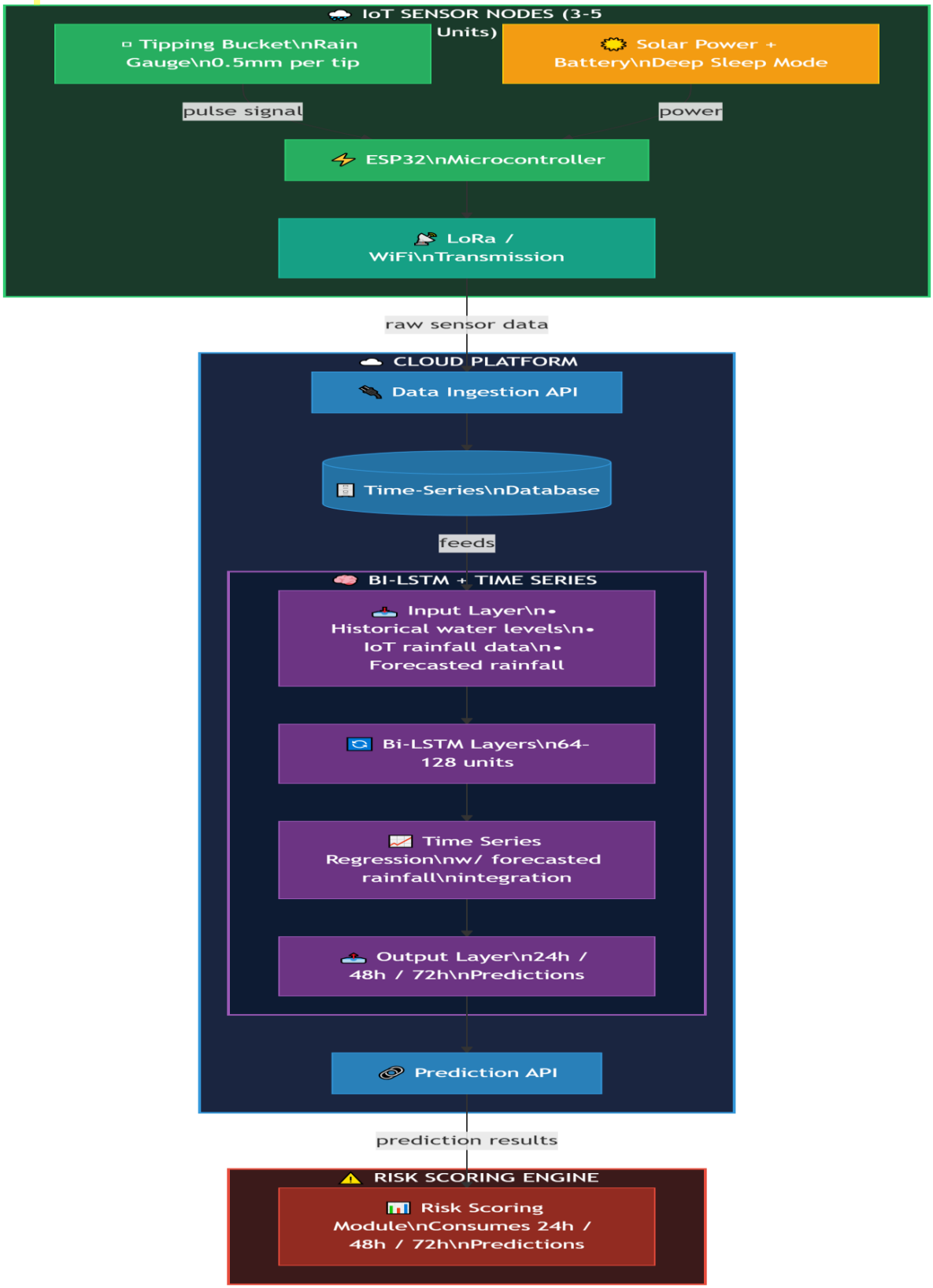


Figure 5 - SYSTEM ARCHITECTURE

Component Details

1. IoT Sensor Network:

- **Hardware:** 3-5 ESP32-based nodes with tipping bucket rain gauges
- **Deployment Locations:** Strategic points along Kelani River basin (upstream, midstream, downstream)
- **Communication:** LoRa for rural areas, WiFi fallback where available
- **Power:** Solar with battery backup for 7-day autonomy
- **Data:** Rainfall intensity (mm/15min), battery voltage, timestamp

2. Data Acquisition Service:

- **IoT Data:** MQTT broker receiving sensor transmissions
- **Government Data:** API clients fetching Irrigation Department river levels (10-min intervals)
- **Historical Data:** Batch imports from archives and CRAN "Ditwah" R package
- **Validation:** Range checks, spike detection, missing data handling

3. Prediction Engine:

- **Model:** Hybrid Bi-LSTM + Time Series Regression
- **Inputs:** Historical water levels (9 days), real-time IoT rainfall, forecasted rainfall
- **Outputs:** Water level predictions for 24h, 48h, 72h ahead
- **Frequency:** Runs every 3 hours, or on-demand via API

4. Prediction API:

- **Endpoints:** RESTful API for model serving
- **Authentication:** API key for authorized access (Student B's components)
- **Documentation:** OpenAPI 3.0 specification

Data Flow

1. **Real-time Flow (15-minute cycle):**
    - IoT sensors measure rainfall → Transmit via LoRa/WiFi → MQTT broker → Validate → Store in time-series database
    - Irrigation API fetches river levels → Store in database
2. **Prediction Flow (3-hour cycle):**
    - Fetch latest 9 days of historical water levels

    - Fetch recent IoT rainfall data
    - Fetch forecasted rainfall from Meteorology API
    - Run Bi-LSTM model → Generate 24/48/72h forecasts
    - Store predictions in database
    - Make predictions available via API
3. **On-demand Flow:**
    - Student B's risk engine calls prediction API
    - API loads latest model and recent data
    - Returns predictions for requested time horizons

Scalability Considerations

- **Database:** Time-series data partitioned by date for efficient querying
- **Model Serving:** Model containerized for horizontal scaling
- **API:** Stateless design enables load balancing
- **Sensor Network:** Architecture supports adding up to 100 nodes

Security Considerations

- **API Authentication:** JWT tokens for prediction API access
- **Data Encryption:** TLS 1.3 for data in transit
- **Device Authentication:** Unique device IDs with pre-shared keys
- **Secure Storage:** Encrypted model artifacts and credentials

5. USER REQUIREMENTS

5.1 Stakeholder identification

Stakeholder	Role in Components	Needs	Interaction Frequency
Irrigation Department	Primary data provider for river levels and historical data	Accurate sensor calibration, reliable API access, data validation feedback	Daily (API calls)
Department of Meteorology	Provider of historical rainfall and forecast data	Data validation, API key management	Daily (API calls)
Student B (Partner)	Consumer of Prediction API	Real-time forecasts, high availability, low latency (<200ms)	Continuous (API calls)
Disaster Management Centre (DMC)	Secondary stakeholder, dashboard user	Model performance metrics, prediction accuracy reports	Weekly (reports)
SLIIT Supervisor	Academic supervisor	Progress updates, milestone reviews, technical guidance	Weekly (meetings)
Field Deployment Team	Installation and maintenance	Sensor specifications, installation guides, maintenance procedures	During deployment

Table 4 - Stakeholder identification

5.2 Functional Requirements

IoT Sensor Hardware Requirements

Rainfall Measurement

The system shall measure rainfall intensity at 15-minute intervals using deployed IoT sensors. This requirement ensures timely capture of precipitation data critical for flood prediction. The tipping bucket mechanism provides 0.2mm resolution, which is the standard for hydrological applications. Rainfall data is accumulated over each 15-minute window before transmission, with intermediate values stored locally in case of communication failure.

Reliable Data Transmission

The system shall transmit sensor data to the cloud server via LoRa or WiFi with greater than 95% reliability. This requirement addresses the challenge of maintaining communication

during adverse weather conditions when cellular networks often fail. The system implements automatic retry mechanisms with exponential backoff for failed transmissions. A store-and-forward capability buffers up to 30 days of data locally on microSD card, ensuring no data loss during extended communication outages.

Solar Power Autonomy
The system shall operate on solar power with 7-day autonomy during cloudy periods. This requirement ensures continuous operation during monsoon seasons when sunlight is limited. The power management system includes a 10W solar panel, 12V 7Ah battery, and PWM charge controller. Deep sleep mode reduces current consumption to 10µA when the battery level drops below 20%, preserving power for essential functions and extending operational life until sunlight returns.

Sensor Calibration
The system shall calibrate sensors against Department of Meteorology reference stations. This requirement ensures measurement accuracy and data consistency across the national network. Calibration checks occur monthly, comparing the tipping bucket readings against manual gauge measurements at co-located reference stations. An auto-correction factor is applied to compensate for systematic biases, with calibration logs maintained for audit purposes.

Data Acquisition Requirements

River Level Data Acquisition
The system shall fetch river water level data from the Irrigation Department API every 10 minutes. This requirement provides real-time water level readings from official gauging stations at Nagalagam Street, Hanwella, and Kithulgala. The API client implements comprehensive error handling including retry logic for failed requests, timeout management, and fallback to historical data when the API is unavailable.

Rainfall Forecast Acquisition
The system shall fetch forecasted rainfall data from the Department of Meteorology API every 6 hours. This requirement supplies the predictive model with future rainfall inputs essential for 24, 48, and 72-hour forecasts. The system manages API authentication securely using environment variables, respects rate limits to prevent throttling, and caches forecast data to reduce unnecessary API calls.

Historical Data Storage
The system shall collect and store 10+ years of historical water level and rainfall data. This requirement establishes the foundation for model training and validation. Data is stored in a time-series database optimized for hydrological queries, partitioned by date for efficient retrieval. The dataset spans from 2015 to 2025, including the critical Cyclone Ditwah event data extracted from the CRAN "Ditwah" R package.

Data Preprocessing
The system shall preprocess data with missing value interpolation and outlier detection. This requirement ensures data quality before model training. Linear interpolation fills gaps of less

than 6 hours, while longer gaps are flagged for manual review. Outliers are detected using the Interquartile Range (IQR) method with thresholds specific to hydrological patterns. Spike detection algorithms identify and flag anomalous readings that may indicate sensor malfunction.

Machine Learning Model Requirements

Water Level Prediction
The system shall predict water levels for 24, 48, and 72-hour horizons. This requirement delivers the core predictive capability using the hybrid Bi-LSTM and Time Series Regression model. The model takes 72 timesteps (9 days) of historical water levels and integrates real-time IoT rainfall data with forecasted rainfall to generate three future predictions. Predictions are calibrated to actual water levels with confidence intervals for uncertainty quantification.

Prediction Accuracy
The system shall achieve prediction accuracy greater than 85% for 24-hour forecasts. This requirement establishes the performance standard for the model. Accuracy is measured as the percentage of predictions falling within 0.5 meters of actual water levels. The model is validated against the 2025 test dataset and specifically against the Cyclone Ditwah extreme event to ensure robustness during severe weather conditions.

Model Retraining
The system shall retrain the model monthly with new data. This requirement ensures the model adapts to changing hydrological patterns, seasonal variations, and potential land use changes in the catchment area. The retraining pipeline automates data collection, preprocessing, model training, validation, and deployment. Model versioning tracks performance across versions, allowing rollback if accuracy degrades.

API Requirements

Prediction API
The system shall provide prediction API with endpoints for single and batch requests. This requirement enables integration with Student B's risk scoring engine and other downstream systems. The API follows RESTful principles, accepting JSON requests and returning JSON responses. Single prediction endpoints accept one station request, while batch endpoints process multiple stations efficiently. All endpoints include comprehensive error handling with meaningful status codes.

API Documentation
The system shall provide API documentation with OpenAPI/Swagger specification. This requirement ensures that Student B and future developers can integrate with the API effectively. The documentation includes endpoint descriptions, request and response schemas, authentication requirements, rate limits, and example requests. Interactive Swagger UI allows testing endpoints directly from the documentation.

Monitoring and Logging Requirements

Data Logging
The system shall log all sensor readings and predictions for audit purposes. This requirement provides traceability and supports post-event analysis. Logs include sensor readings with timestamps, API requests and responses, model predictions with input features, and system events. Log retention is 30 days for raw data with aggregated statistics retained for 10 years. Export capability allows downloading logs for external analysis.

Health Monitoring
The system shall send health status alerts for sensor nodes. This requirement enables proactive maintenance and ensures system reliability. Alerts are triggered for low battery (below 20%), communication failure (missing data for 4 hours), sensor malfunction (unusual readings), and enclosure tampering. Notifications are delivered via email and SMS to the designated maintenance team, with priority levels based on severity.

5.3 Non-Functional Requirements

Performance Requirements

Model Inference Time
The system shall complete model inference in less than 30 seconds for each prediction request. This requirement ensures timely delivery of forecasts to downstream systems. The optimized model architecture uses quantized weights and efficient tensor operations to minimize inference time. GPU acceleration is utilized when available, with CPU fallback for cost-constrained deployments.

API Response Time
The API shall respond within 200 milliseconds for 95% of requests (P95). This requirement supports real-time integration with risk scoring and alert systems. Response time includes network latency, request processing, model inference, and response serialization. Caching of recent predictions reduces latency for frequently requested horizons.

Data Transmission Latency
Sensor data shall be available in the cloud within 2 minutes of measurement. This requirement ensures the prediction model receives the most current rainfall information. Transmission latency includes sensor processing time, communication channel delay, and cloud ingestion processing. During normal conditions, latency averages 30 seconds; during heavy rainfall events, event-based transmission reduces latency to under 10 seconds.

Training Time
Full model retraining shall complete within 4 hours. This requirement enables monthly retraining cycles without excessive computational overhead. Training time includes data loading, preprocessing, model training with early stopping, validation, and deployment. Distributed training across multiple GPU instances can reduce time to under 1 hour when needed.

Concurrent Request Handling
The API shall handle at least 100 concurrent requests per second. This requirement ensures the system remains responsive during flood events when alert volumes spike. Horizontal scaling through container orchestration automatically provisions additional instances when load exceeds thresholds. Load balancing distributes requests across available instances.

Query Performance
Database queries for 30 days of data shall execute in under 500 milliseconds. This requirement supports real-time dashboard visualizations and ad-hoc analysis. Time-series database optimizations include time partitioning, columnar storage for metrics, and materialized views for common aggregations. Indexes are maintained on frequently queried fields.

Reliability Requirements

Sensor Uptime
Individual sensor nodes shall maintain uptime greater than 98%. This requirement ensures continuous data collection from the hyperlocal network. Uptime is measured from deployment, excluding planned maintenance windows. Redundant components and remote monitoring enable quick identification and resolution of issues. Watchdog timers reset the system automatically in case of firmware hangs.

Transmission Success Rate
Data transmission success rate shall exceed 95% over the operational period. This requirement ensures reliable data delivery to the cloud platform. Success is measured by delivery confirmation from the cloud broker. Failed transmissions trigger automatic retries with exponential backoff, up to 5 attempts. Persistent failures after 24 hours trigger maintenance alerts.

API Uptime
The prediction API shall maintain 99.5% uptime excluding planned maintenance. This requirement ensures continuous availability for Student B's risk scoring engine. Uptime is measured across all API endpoints, with maintenance windows scheduled during periods of low expected demand (early morning hours). Redundant deployments across multiple availability zones provide failover capability.

Data Retention
Raw sensor data shall be retained for 30 days, with aggregated data retained for 10 years. This requirement balances storage costs against analytical needs. Data retention policies are enforced through automated archiving and deletion processes. Aggregated data includes daily and monthly summaries, flood event snapshots, and model training datasets.

Power Autonomy
Each sensor node shall operate for 7 days without sunlight. This requirement ensures continuous operation during extended cloudy periods common in monsoon seasons. Power autonomy is measured from full battery charge with the solar panel disconnected. The power

budget calculation accounts for worst-case current consumption including deep sleep, sensing, and transmission activities.

Model Availability
The prediction model shall be available for inference 99% of the time. This requirement ensures forecasts are always available when needed. Model availability includes the serving infrastructure, model loading, and inference endpoints. Multiple model versions can run simultaneously to enable blue-green deployments without downtime.

Security Requirements

API Authentication
All API endpoints shall require authentication. This requirement prevents unauthorized access to prediction services and protects system resources. Authentication uses JWT tokens with expiration and refresh mechanisms. API keys are issued per client (Student B, DMC) with individual quotas and permissions.

Data Encryption in Transit
All data transmitted over networks shall be encrypted using TLS 1.3. This requirement protects sensitive data during transmission between sensors, cloud services, and API clients. HTTPS is enforced for all web endpoints. LoRa transmissions are encrypted using AES-128 with network keys to prevent eavesdropping.

Device Authentication
Each sensor node shall authenticate with the cloud platform using unique credentials. This requirement prevents unauthorized devices from injecting false data. Pre-shared keys are stored in secure element memory on the ESP32. Device certificates provide mutual TLS authentication for cloud connections.

Rate Limiting
The API shall enforce rate limits of 1000 requests per minute per API key. This requirement prevents abuse and ensures fair resource allocation. Rate limits are enforced at the API gateway level using token bucket algorithm. Exceeded limits return 429 status codes with retry-after headers.

Audit Logging
All API access and data modifications shall be logged for audit purposes. This requirement provides traceability for security investigations. Logs include timestamp, client identifier, endpoint accessed, request parameters, response status, and source IP address. Logs are immutable once written and accessible only to authorized personnel.

Credential Storage
Credentials shall be stored encrypted using AES-256. This requirement protects sensitive information such as API keys, database passwords, and device secrets. Credentials are never stored in source code or exposed in logs. Environment variables and secrets management services provide secure injection at runtime.

Usability Requirements

API Documentation
API documentation shall be complete and include request/response examples. This requirement enables efficient integration by Student B and future users. Documentation follows OpenAPI 3.0 specification with interactive Swagger UI for testing. Examples cover success responses, error cases, and edge conditions.

Model Interpretability
The API shall provide feature importance information with predictions. This requirement enables stakeholders to understand what factors influenced each forecast. Feature importance is calculated using SHAP (SHapley Additive exPlanations) values for the regression branch. Importance values help explain why a prediction changed from previous forecasts.

Error Messages
Error messages shall be clear, actionable, and include reference codes. This requirement helps developers quickly diagnose and resolve issues. Error responses include a machine-readable code, human-readable message, and suggested resolution. Documentation provides detailed explanations for each error code.

Deployment Documentation
Sensor deployment shall include comprehensive installation guides. This requirement ensures field teams can correctly install and configure sensors. Documentation includes site selection criteria, mounting instructions, wiring diagrams, and calibration procedures. Troubleshooting guides address common issues and maintenance procedures.

Maintainability Requirements

Modular Architecture
The system shall follow modular architecture with decoupled components. This requirement enables independent development, testing, and deployment of each component. Microservices design separates IoT ingestion, data processing, model training, model serving, and API layers. Components communicate through well-defined APIs with versioning.

Code Documentation
Source code shall maintain documentation coverage greater than 70%. This requirement ensures future maintainers can understand and modify the codebase. Documentation includes function docstrings, module-level explanations, and architectural decisions. Critical algorithms include mathematical formulas and references to research papers.

Version Control
All code shall be managed through Git with meaningful commit history. This requirement provides traceability and collaboration capabilities. The repository follows standard branching strategy with main for production, develop for integration, and feature branches for development. Commit messages follow conventional format with types and descriptions.

CI/CD Pipeline
Continuous integration and deployment shall be automated through GitHub Actions. This requirement ensures quality and reliability of deployments. CI pipeline runs unit tests, linting,

and security scanning on every commit. CD pipeline deploys to staging environment for testing and production after approval.

Remote Firmware Updates
Sensor nodes shall support over-the-air (OTA) firmware updates. This requirement enables remote deployment of bug fixes and feature improvements without physical access. OTA updates use ESP32's built-in capabilities with secure update verification. Failed updates roll back to previous working version automatically.

Scalability Requirements

Sensor Capacity
The system shall support up to 100 sensor nodes. This requirement accommodates expansion to the full Kelani River network and potential extension to other basins. Horizontal scaling of ingestion pipelines handles increased data volume. Database partitioning strategies prevent performance degradation.

API Scalability
The API shall auto-scale to handle traffic spikes during flood events. This requirement ensures service availability when demand increases during emergencies. Kubernetes-based orchestration monitors CPU and memory utilization, automatically adding instances when thresholds are exceeded. Load balancers distribute traffic across available instances.

Database Scalability
The database shall handle up to 1 terabyte of time-series data. This requirement supports long-term data retention and expansion to additional sensors. TimescaleDB partitioning automatically manages data distribution across storage tiers. Hot data (recent 30 days) resides on fast SSD storage, while cold data moves to cost-effective object storage.

Model Versioning
The system shall support hot-swappable model versions. This requirement enables zero-downtime updates when new models are deployed. Model registry tracks performance metrics for each version. Canary deployments route a percentage of traffic to new models for validation before full rollout.

6. COMMERCIALIZATION PLAN

Target Market & Customer Persona

Market Segment	Key Customers	Estimated Size
Primary	Irrigation Department, DMC, Meteorology Dept	15 river basins, 50+ gauging stations
Secondary	UNDP, World Bank, Red Cross	10+ flood-prone countries in South Asia
Tertiary	Research institutions, universities	20+ institutions in Sri Lanka

Table 5 - Target Market & Customer Persona

Value Proposition

For Government Agencies	For Disaster Management	For International Partners
0.96 R^2 accuracy (16% improvement)	3-6 hours additional warning time	Low-cost (70% cheaper than imports)
10-minute updates vs. 6-12 hour delays	72-hour forecasts for proactive planning	Scalable to other river basins
LKR 18,200 per node vs. LKR 60,000+	Data-driven evacuation priorities	Validated on Cyclone Ditwah

Table 6 - Value Proposition

Revenue Model

Revenue Stream	Target	Price (LKR)	Annual Potential
IoT Sensor Node Sales	Irrigation Dept	25,000 - 35,000 per node	500,000 - 2,000,000
Prediction API Subscription	DMC, Student B	50,000/month or 500,000/year	500,000 - 1,500,000
Dashboard License	DMC, Local Govt	100,000 - 300,000/year	500,000 - 1,500,000
Training & Support	All stakeholders	50,000 - 100,000/workshop	200,000 - 500,000

Total Annual Potential		1.7M - 5.5M

Table 7 - Revenue Model

Cost Estimation

Category	**Total (LKR)**	**Justification**
Hardware (5 nodes + spares)	160,600	ESP32, rain gauge, LoRa, solar, battery
Tools	16,000	Multimeter, soldering station
Cloud Services (12 months)	132,000	AWS/GCP hosting, API, database
Field Work	45,000	Transportation, calibration
Contingency (10%)	35,360	Unforeseen expenses
TOTAL	389,960	

Table 8 - Cost Estimation

Competitive Advantage

Competitor	**Weakness**	**Our Advantage**
International Systems (Delft-FEWS)	Expensive (LKR 5M+), foreign expertise	Low cost (LKR 400k), locally developed
IWMI AWARE	Macro-level only, no IoT	Household-level, real-time sensors
DMC Current System	Manual gauges, 12-hour delays	Automated, 10-minute updates
Academic Prototypes	Not operational, siloed	End-to-end, API-ready

Table 9 - Competitive Advantage

BUDGET AND JUSTIFICATION

Total Project Budget

Category	Item	Qty	Unit Price (LKR)	Total (LKR)
HARDWARE - PER NODE				
	ESP32 Development Board	1	1,500	1,500
	Tipping Bucket Rain Gauge	1	5,000	5,000
	Hall Effect Sensor (A3144)	1	200	200
	LoRa Module (RFM95W)	1	2,000	2,000
	Solar Panel (10W Polycrystalline)	1	3,000	3,000
	Battery (12V 7Ah Lead-Acid)	1	2,500	2,500
	PWM Charge Controller (10A)	1	1,200	1,200
	IP67 Weatherproof Enclosure	1	1,500	1,500
	Mounting Hardware (Pole, Brackets)	1 set	1,000	1,000
	MicroSD Card (16GB)	1	500	500
	Cost Per Complete Node			18,400
HARDWARE - TOTAL (5 NODES)				
	5 Complete Sensor Nodes	5	18,400	92,000
	Spare Components (ESP32, LoRa, Sensors)	1 set	6,000	6,000
	Hardware Subtotal			98,000

SOFTWARE & CLOUD				
	Cloud Hosting (AWS/GCP) - 12 months	1	80,000	80,000
	Domain Name Registration	1	2,000	2,000
	SSL Certificate	1	0	0
	API Access (Meteorology Dept)	1	5,000	5,000
	Software Subtotal			87,000
TOOLS & FIELD WORK				
	Multimeter	1	5,000	5,000
	Soldering Station	1	0	0
	Transportation (Deployment Visits)	4 trips	5,000	20,000
	Calibration Equipment (Reference Gauge)	1 set	10,000	10,000
	Field Assistant	2 days	5,000	10,000
	Tools & Field Work Subtotal			45,000
	TOTAL DIRECT COSTS			230,000
	Contingency (10%)			23,000
	GRAND TOTAL			253,000

Table 10 - Total Project Budget

Budget Justification

Hardware (LKR 98,000)

Each complete sensor node costs LKR 18,400. Five nodes will be deployed at strategic locations along the Kelani River basin (upstream at Kithulgala, midstream at Hanwella and Glencourse, downstream at Nagalagam Street and Ambatale). An additional set of spare components (ESP32, LoRa module, Hall effect sensors) is included for field replacements. The tipping bucket rain gauge provides 0.2mm resolution, meeting hydrological standards. LoRa modules enable 5-8km communication range where cellular networks may fail during floods. The solar power system with battery provides 7-day autonomy during cloudy periods.

Software & Cloud (LKR 87,000)

Cloud hosting costs are reduced using AWS Free Tier which provides 750 hours per month of EC2 and RDS services for 12 months at no cost, with the budget covering only potential overages. Domain registration ensures professional API access. Free SSL certificates from Let's Encrypt provide secure communication. Meteorology Department API access is mandatory for obtaining forecasted rainfall data essential for model predictions.

Tools & Field Work (LKR 45,000)

The multimeter is essential for testing circuits and sensor calibration. The soldering station is borrowed from the university lab to reduce costs. Transportation covers four field trips to Kelani River deployment locations for site selection, installation, calibration, and maintenance. Calibration equipment ensures sensor accuracy against reference standards. A field assistant helps with installation at remote riverbank locations.

Contingency (LKR 23,000)

A 10% contingency covers unexpected expenses such as component price fluctuations, shipping delays requiring expedited delivery, or component failures during testing requiring replacement.

8. WORK BREAKDOWN STRUCTURE (WBS)

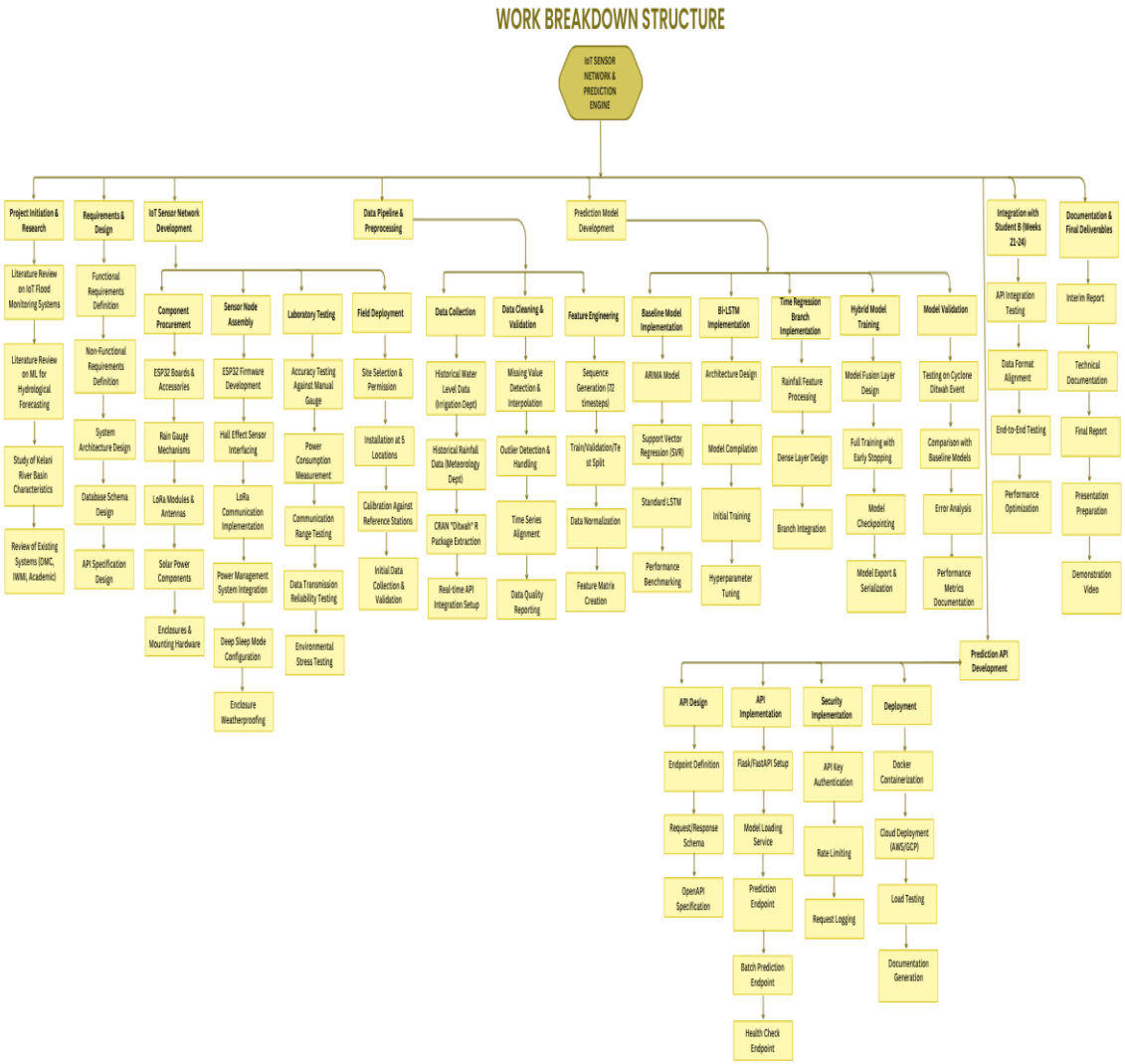


Figure 6 -WORK BREAKDOWN STRUCTURE

9. GANTT CHART

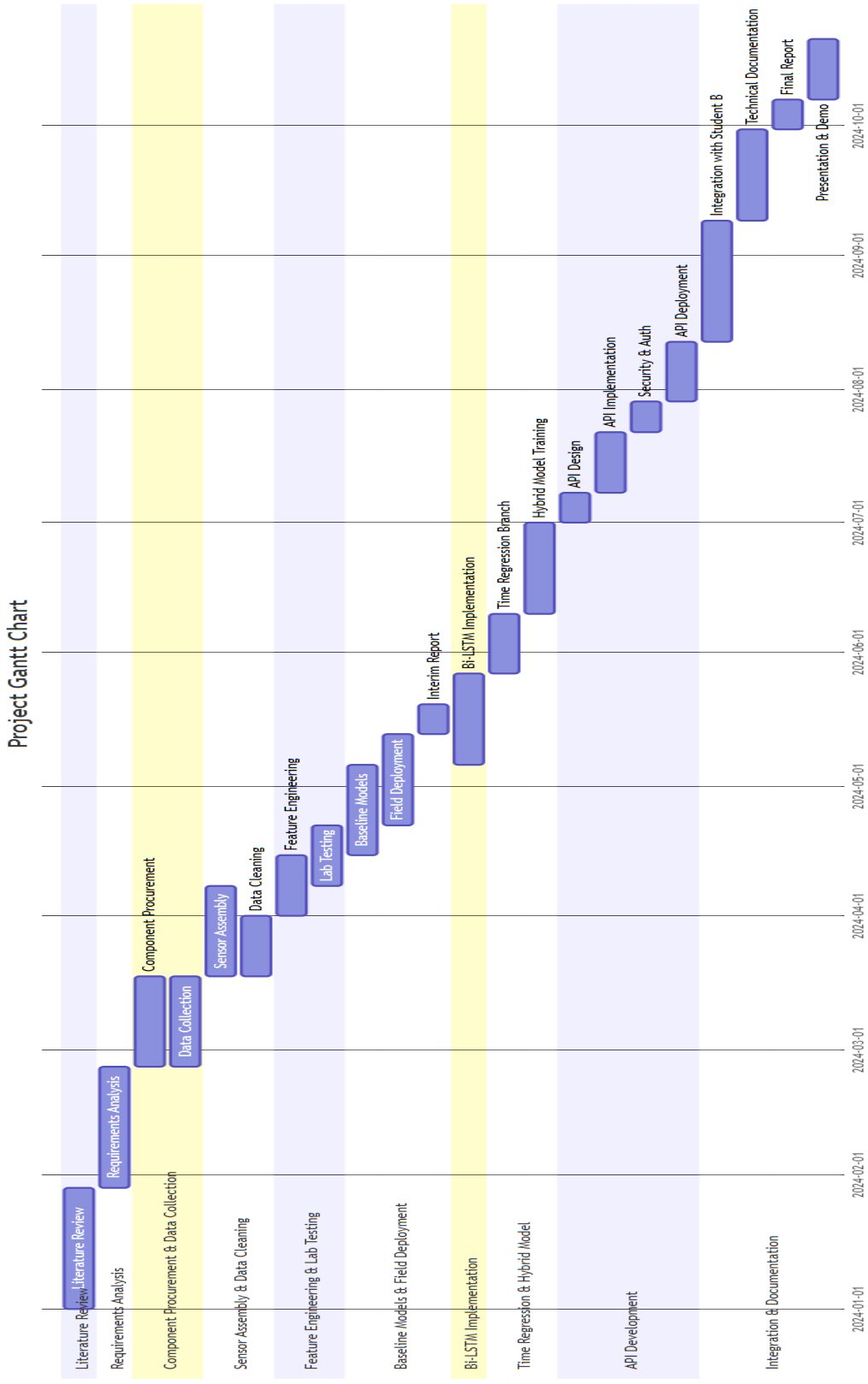


Figure 7 -GANTT CHART

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